Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Bengaluru, India

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Summary

Cloud & DevOps/MLOps Engineer with 2 years at AWS Professional Services as ProServe (Cloud Consultant) - DevOps/MLOps, delivering DevOps and cloud-platform engagements for financial-services and technology customers (RWS, State Street, DTCC). Fluent across AWS landing zones and governance (Organizations, SCPs, Terraform, GitHub Actions), production MLOps on SageMaker (pipelines, blue-green endpoints, drift detection), full-stack apps (React, FastAPI, Node.js), and AI-assisted developer tooling (Claude Code plugins, MCP servers, agent skills). Author of 2 samples published by AWS (aws-samples). 11 upstream PRs merged, including axios, Apache Airflow, lucide, and multer. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems, 107 contests.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Bengaluru, India

- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Cut per-account setup from 3+ days to under 4 hours (~90%) by replacing a legacy account-vending pipeline with Terraform Cloud (OIDC) for a data-centre exit and a second landing zone: 4,000+ resources across 20+ workspaces, none failing.
  - Codified Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines with a security baseline (IMDSv2, EBS encryption, no default VPC); every change ran in a sandbox mirror first.
  - Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service onboarding: route table within 15 minutes, firewall access by tag in about a minute. Used Kiro and Claude Code for specs and design documentation.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC with Amazon Q.
- **MLOps Pipeline (SME Program, Jun – Aug 2025):** Built a SageMaker medical-image-classification pipeline that trains, gates and ships models, with blue/green canary endpoints, auto-scaling, and PSI drift detection, provisioned via Terraform + CodePipeline behind CloudFront and API Gateway + Lambda. Generalized and published by AWS as an official sample (aws-samples, MIT-0).
- **Internal Impact:** Published a second AWS sample on Organizations governance (see Publications). Co-authored an AWS Prescriptive Guidance pattern on secure PR previews and approval-gated releases for Amplify Gen 2. MLOps SME Program Graduate. 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Delivered 2 org-wide tech talks on Terraform testing (2025) and GitOps pipeline gating (2026). Associate Speaker Certified (2026).

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated an on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – enabling zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized a production application on AWS ECS/ECR with GitHub Actions CI/CD, and deployed SonarQube on EC2 + RDS to embed continuous code-quality analysis into the delivery pipeline.

Publications & Recognition

- **Published by AWS:** aws-samples/sample-sagemaker-image-classification-mlops (author, MIT-0) – SageMaker MLOps in 86 Terraform files across 4 stacks and 13 modules: clinical quality and Fairlearn fairness gates, hourly PSI drift that starts retraining, canary deploys with auto-rollback. *2026*
- **Published by AWS:** aws-samples/sample-aws-terraform-org-governance (author, MIT-0) – Terraform for AWS Organizations governance on Control Tower: six policy types as 20 policy documents, delegated admin for 15 services. *2026*
- **LeetCode Guardian** (top 1.1%) – 2166 peak rating, 1300+ problems, 107 contests. GeeksforGeeks 500+ problems; HackerRank 6-Star Problem Solving.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Personal Projects

Claude Cost Optimizer Mar 2026
Claude Code Plugin, MCP, TypeScript, React, Python GitHub | Live
◇ Claude Code plugin, MCP cost server, CLI grader and web calculator for Claude spend. Rebuilt every estimator on cache-aware billing, correcting a modelled 30-turn Opus session from $3.44 to $0.74, and replaced every vague savings range with a sourced or computed number. Golden-value tests pin the cost math in all three implementations.

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, NumPy, FastAPI, React, Docker GitHub | Live
◇ Rating model on 15 engineered features. Found and fixed a train/serve skew on participant count that had tripled backtest error (MAE 38.76 to 12.46 on 57 contests). Moved serving to NumPy-only inference pinned by a golden test, cutting the runtime from ~1.8 GB to ~65 MB; a weekly job refreshes the data and re-runs the backtest.

Blue-Green AWS Terraform Jul 2025
Terraform, ECS on EC2, ALB, CodePipeline, CodeDeploy GitHub
◇ 15 Terraform modules for ECS blue/green on an EC2 Auto Scaling Group: 30% canary for 1 minute with rollback on a 5xx spike or unhealthy target, an 80% coverage gate in CodeBuild, and 8 `terraform test` files on a mock provider, plus Checkov and tflint on every PR.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API GitHub
◇ MCP server putting 26 text models from 7 providers, plus image, video and embedding models, behind 8 tools and one set of AWS credentials. Benchmarked in us-east-1: 26 of 26 models answered and 156 of 156 calls succeeded.

itr-agent Jul 2026
TypeScript, MCP Protocol, Node.js, Zod GitHub
◇ Local-first MCP server for Indian income tax: the LLM interviews and calls 12 tools, while a deterministic engine over a versioned JSON rule pack computes every rupee. 151 tests pin it to published worked examples and the statute, on an Ubuntu and Windows CI matrix.

Other Notable Projects: Ledger Sync (self-hosted finance app, 26 pages, AI assistant), GitScope (Chrome Web Store extension), skillcheck (Agent Skills linter, 284 npm downloads a month), Kinfolk (collaborative family-graph app on Next.js + Neon), Instagram Likes Leaderboard (browser-console tool).

Open Source Contributions (11 merged upstream)

- **axios** (109K☆) – AxiosError message now built from AggregateError sub-errors, ending blank dual-stack errors (#11059)
- **Apache Airflow** (47K☆) – template_fields on SalesforceBulkOperator, so its arguments render via Jinja at run time (#63109)
- **lucide** (25K☆) – Fixed a TypeError crash when Vue templates pass the kebab-case icon-node prop (#4512)
- **multer** (12K☆) – Files rejected by fileFilter no longer use up maxCount and fail later valid uploads (#1426)
- **terraform-aws-modules/atlantis** – Fixed custom ECS security group egress rules being silently replaced by allow-all (#446)
- **Also merged:** PrefectHQ/prefect, feast-dev/feast, awslabs/agent-plugins (#132, #212), awslabs/mcp, cloudposse/terraform-aws-tfstate-backend; co-author of the aws-ia/terraform-aws-vpc v4.8.0 fix.

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Technical Skills

Cloud & Infrastructure: AWS (Organizations, SCPs, Landing Zone, Control Tower, Transit Gateway, Direct Connect, VPN, Security Hub, Config, IAM, ECS, Lambda, API Gateway, CloudFront, EventBridge, CloudWatch, X-Ray, CodePipeline, CodeDeploy), Terraform, Docker, Ansible
DevOps & Automation: GitHub Actions, CI/CD, GitOps, Blue-Green Deployments, tftest, PyTest, cfn-lint/cfn-guard, SonarQube
MLOps & AI/ML: SageMaker (Pipelines, Endpoints, Custom PSI drift jobs), TensorFlow, PyTorch, Scikit-learn, RAG Pipelines, LLMOps
AI Developer Tools: Claude Code (custom skills, MCP servers, hooks, subagents), Amazon Q, Kiro, GitHub Copilot, MCP Protocol, AWS Bedrock (Nova, Claude, Llama)
Languages: Python, HCL, TypeScript, JavaScript, C++, Java, SQL, Bash
Frameworks & Data: React, Next.js, Node.js, Express, FastAPI, MongoDB, PostgreSQL, SQLite